ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 8 · SET 01

Max. Marks: 60 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The smallest 4-digit perfect square number is:

(a) 1000 (b) 1089 (c) 1024 (d) 1156

Ans: (c) $32^2 = 1024$, and $31^2 = 961$ has only 3 digits.

(ii) The cube root of 17576 is:

(a) 24 (b) 28 (c) 26 (d) 36

Ans: (c) $17576 = 2^3 \times 13^3 \Rightarrow \sqrt[3]{17576} = 2 \times 13 = 26$.

(iii) The standard form of 0.00045 is:

(a) 4.5×10^{-3} (b) 45×10^{-5} (c) 4.5×10^{4} (d) 4.5×10^{-4}

Ans: (d) Move the decimal 4 places to the right: 4.5×10^{-4} (45×10^{-5} is not in standard form).

(iv) The Roman numeral XCIV represents:

(a) 86 (b) 94 (c) 106 (d) 114

Ans: (b) XC = 90 and IV = 4, so XCIV = 94.

(v) The binary number 1011 is equal to which decimal number?

(a) 11 (b) 9 (c) 10 (d) 13

Ans: (a) $1\times8 + 0\times4 + 1\times2 + 1\times1 = 11$.

(vi) Which quadrilateral always has equal diagonals that bisect each other, but the diagonals need not be perpendicular?

(a) Rhombus (b) Kite (c) Rectangle (d) Parallelogram

Ans: (c) Rectangle: diagonals equal and bisect each other; they are perpendicular only if it is a square.

(vii) A number is divisible by 18 if it is divisible by:

(a) 2 and 9 (b) 3 and 6 (c) 4 and 9 (d) 2 and 6

Ans: (a) 18 = 2 × 9 with 2 and 9 co-prime. (3 and 6 are not co-prime, so they do not guarantee divisibility by 18.)

(viii) The value of $104^2 - 96^2$ is:

(a) 800 (b) 1600 (c) 1200 (d) 16

Ans: (b) $(104 + 96)(104 - 96) = 200 \times 8 = 1600$.

(ix) In the expansion of $(7x - 3)^2$, the middle term is:

(a) −42x (b) −21x (c) +42x (d) −9x

Ans: (a) $(7x - 3)^2 = 49x^2 - 2(7x)(3) + 9 = 49x^2 - 42x + 9$.

(x) Which of the following pairs of ratios forms a proportion?

(a) 15 : 25 and 9 : 15 (b) 4 : 6 and 9 : 12

(c) 3 : 5 and 5 : 3 (d) 2 : 7 and 7 : 2

Ans: (a) 15/25 = 3/5 and 9/15 = 3/5, so 15 : 25 :: 9 : 15.

(xi) The ratio of 45 minutes to 1½ hours in simplest form is:

(a) 3 : 2 (b) 1 : 2 (c) 1 : 3 (d) 2 : 3

Ans: (b) 1½ h = 90 min, so 45 : 90 = 1 : 2.

(xii) Sum of any two consecutive odd numbers is always divisible by:

(a) 3 (b) 4 (c) 5 (d) 6

Ans: (b) $(2n-1)+(2n+1) = 4n$, divisible by 4.

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the square root of 2916 by prime factorisation.

$2916 = 2^2 \times 3^6$ (1 mark)
$\sqrt{2916} = 2 \times 3^3 = 54$ (1 mark)

3. Write 0.000000837 in standard form.

Move the decimal 7 places right: 8.37 (1 mark)
$0.000000837 = 8.37 \times 10^{-7}$ (1 mark)

4. Write 2024 in Roman numerals.

2000 = MM, 20 = XX, 4 = IV (1 mark)
2024 = MMXXIV (1 mark)

5. Convert the decimal number 45 into base 2.

45 ÷ 2 → remainders 1,0,1,1,0,1 (1 mark)
Reading upwards: $45 = (101101)_2$ (1 mark)

OR

Find the decimal value of (2314) in base 5.

$2\times5^3 + 3\times5^2 + 1\times5 + 4 = 250 + 75 + 5 + 4$ (1 mark)
= 334 (1 mark)

6. Two adjacent angles of a parallelogram are in the ratio 2 : 3. Find all the angles of the parallelogram.

Adjacent angles are supplementary: $2x + 3x = 180° \Rightarrow x = 36°$ (1 mark)
Angles: 72°, 108°, 72°, 108° (1 mark)

7. Find x if x : 12 :: 15 : 36.

Product of extremes = product of means: $36x = 12 \times 15$ (1 mark)
$x = 180/36 = 5$ (1 mark)

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Find the sum without converting to Hindu-Arabic numerals:
(i) XLVII + LXIII (ii) CCXLIII + DCLVII

(i) XLVII + LXIII → (XL + VII) + (LX + III): XL + LX = C and VII + III = X, so the sum is CX [47 + 63 = 110] (2 marks)
(ii) CCXLIII + DCLVII: 243 + 657 = 900 = CM (1 mark)

9. The diagonals of a rhombus are 16 cm and 30 cm. Find the length of its side and its perimeter.

Diagonals bisect at right angles: half-diagonals 8 cm and 15 cm (1 mark)
Side = $\sqrt{(8^2 + 15^2)} = \sqrt{289} = 17$ cm (1 mark)
Perimeter = $4 \times 17 = 68$ cm (1 mark)

OR

In parallelogram PQRS, $\angle P = (3x + 10)°$ and $\angle Q = (2x - 10)°$. Find x and all the angles.

Adjacent angles are supplementary: $3x + 10 + 2x - 10 = 180 \Rightarrow x = 36$ (1 mark)
$\angle P = 118°$, $\angle Q = 62°$ (1 mark)
$\angle R = 118°$, $\angle S = 62°$ (opposite angles equal) (1 mark)

10. Expand and simplify:
(i) $(x + 4)(x^2 - 2x + 3)$ (ii) $(2m - 1)(m + 3)$

(i) $x^3 - 2x^2 + 3x + 4x^2 - 8x + 12 = x^3 + 2x^2 - 5x + 12$ (2 marks)
(ii) $2m^2 + 6m - m - 3 = 2m^2 + 5m - 3$ (1 mark)

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 5292 must be multiplied so that the product is a perfect square. Also find the square root of the product.
(ii) Find the side of a square whose area is 1296 m^2.

(i) $5292 = 2^2 \times 3^3 \times 7^2$ (1 mark); 3 is unpaired, so multiply by 3 (1 mark)
Product = $15876 = 2^2 \times 3^4 \times 7^2$, $\sqrt{} = 2 \times 9 \times 7 = 126$ (1 mark)
(ii) side = $\sqrt{1296} = 36$ m (2 marks: $2^2 \times 2^2 \times 3^4 \rightarrow 2 \cdot 2 \cdot 9 = 36$)

OR

(i) Find the smallest number by which 3087 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient.
(ii) The volume of a cube is 5832 cm^3. Find the length of its edge.

(i) $3087 = 3^2 \times 7^3$ (1 mark); divide by $3^2 = 9$ (1 mark)
Quotient $343 = 7^3$, cube root = 7 (1 mark)
(ii) $5832 = 2^3 \times 3^6$ (1 mark); edge $= 2 \times 3^2 = 18$ cm (1 mark)

12. If $13^2 = 169$, find the values of: (i) 1.3^2 (ii) 0.13^2 (iii) 130^2 (iv) 1300^2 (v) 0.013^2

(i) 1.69 (ii) 0.0169 (iii) 16900 (iv) 1690000 (v) 0.000169 (1 mark each; place the decimal using 2n places)

OR

Simplify and write in exponential form: (i) $7^{-3} \times 7^{-2}$ (ii) $3^9 \div 3^5$ (iii) $(5^{-2})^{-3}$ (iv) $2^7 \div 2^{-3}$ (v) $a^4b^6 \times (ab)^3$

(i) 7^{-5} (ii) 3^4 (iii) 5^6 (iv) 2^{1^0} (v) a^7b^9 (1 mark each)

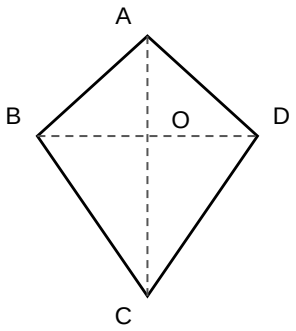
13. (i) Find the digit(s) that can replace * in 52*8 so that the number is divisible by 3.
(ii) Is 3141592 divisible by 11? Show your work.
(iii) Find the smallest 3-digit number divisible by 4 and 9 both.
(iv) Replace the blank to make 7_6 divisible by 4 (smallest digit).
(v) A number is divisible by 6. Name its two divisibility conditions.

(i) 5+2+*+8 = 15 + *, so * = 0, 3, 6, 9 (1 mark)
(ii) (3+4+5+2) − (1+1+9) = 14 − 11 = 3, not 0 or a multiple of 11 ⇒ NOT divisible (1 mark)
(iii) LCM(4,9) = 36; smallest 3-digit multiple = 108 (1 mark)
(iv) last two digits _6 must be divisible by 4: 16 → digit 1 (1 mark)
(v) divisible by 2 and by 3 (1 mark)

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. Riya and Aarav are making a kite for the school exhibition. The kite ABCD has two pairs of adjacent equal sides: AB = AD = 6 cm and CB = CD = 9 cm. Its diagonals AC and BD meet at O (see figure).



(i) What is the length of the side CD?

> CD = CB = 9 cm

(ii) Name the pair of adjacent angles of the kite that are equal, and state the reason.

> ∠B = ∠D (angles between the unequal sides; ΔABC ≅ ΔADC by SSS) — accept ∠ABC = ∠ADC

(iii) Show that the diagonal AC bisects ∠A and ∠C.

> In ΔABC and ΔADC: AB = AD, CB = CD, AC common ⇒ ΔABC ≅ ΔADC (SSS) (1 mark)
> So ∠BAC = ∠DAC and ∠BCA = ∠DCA (CPCT) ⇒ AC bisects ∠A and ∠C (1 mark)

OR

(iii) If ∠AOB = x°, find x and give a reason.

> In a kite the diagonals intersect at right angles (1 mark) ⇒ x = 90° (1 mark)

15. Aditi and Rahul prepare for a Mental Maths Quiz. Their teacher shows the identities: (A) $(a + b)^2 = a^2 + 2ab + b^2$; (B) $(a - b)^2 = a^2 - 2ab + b^2$; (C) $a^2 - b^2 = (a + b)(a - b)$.

(i) Find 53^2 using identity (A).

> $(50 + 3)^2 = 2500 + 300 + 9 = 2809$

(ii) Find 1.02×0.98 using identity (C).

> $(1 + 0.02)(1 - 0.02) = 1 - 0.0004 = 0.9996$

(iii) Prove that $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$.

> LHS = $x^2 + 2xy + y^2 + x^2 - 2xy + y^2$ (1 mark) = $2x^2 + 2y^2 = 2(x^2 + y^2)$ = RHS (1 mark)

OR

(iii) Find $1003^2 - 997^2$ using a suitable identity.

> $(1003 + 997)(1003 - 997)$ (1 mark) = $2000 \times 6 = 12000$ (1 mark)

16. The school garden club prepares a fertiliser mixture of urea, potash and compost in the ratio 2 : 3 : 5. The total weight of the mixture is 40 kg.

(i) How many kilograms of compost are in the mixture?

> Total parts = 10; compost = 5/10 × 40 = 20 kg

(ii) Write the ratio of urea to compost in simplest form.

> 2 : 5

(iii) The club uses 7.5 kg of potash in another batch of the same mixture. What is the total weight of that batch?

> 3 parts = 7.5 kg ⇒ 1 part = 2.5 kg (1 mark); total = 10 × 2.5 = 25 kg (1 mark)

OR

(iii) Find the missing numbers so that these ratios are proportional to 2 : 5: 14 : __ and __ : 45

> 14 : 35 (1 mark) and 18 : 45 (1 mark)